

Statistical Modeling with Python

Concept Reference Notes

1. Hypothesis Testing

Hypothesis testing is used to check whether there is enough evidence to support a claim about data.

We always start with two hypotheses:

- **H_0 (Null Hypothesis):** Usually says there is **no difference or no effect**.
- **H_1 (Alternative Hypothesis):** Says there is **a difference or effect**.

We use $\alpha = 0.05$ as the significance level.

After performing a test, we get a **p-value**.

- If $p < 0.05 \rightarrow$ Reject H_0
- If $p \geq 0.05 \rightarrow$ Fail to reject H_0

Example:

We want to check whether smokers and non-smokers have different medical charges.

- H_0 : There is no difference in charges.
- H_1 : There is a difference in charges.

If $p = 0.001$, then $0.001 < 0.05$, so we **reject H_0** .

This means there is enough evidence to say that the groups are significantly different.

Remember: Fail to reject H_0 does NOT mean H_0 is proved true. It only means we don't have enough evidence to reject it.

2. p-value

The p-value tells us how much evidence we have against H_0 .

Think of it simply:

Small p-value $\rightarrow$ strong evidence against H_0

Large p-value $\rightarrow$ not enough evidence against H_0

For $\alpha = 0.05$:

p-value Decision

$p < 0.05$ Reject H_0

$p \geq 0.05$ Fail to reject H_0

For example:

- $p = 0.02 \rightarrow$ reject H_0
- $p = 0.50 \rightarrow$ fail to reject H_0
- $p = 1.00 \rightarrow$ fail to reject H_0 because 1 is much greater than 0.05

A p-value of 1 does **not** mean "100% probability that H_0 is true."

3. Shapiro-Wilk Test

Shapiro-Wilk is used to check **normality**.

In simple words:

It checks whether the data looks normally distributed.

Hypotheses:

- **H_0 :** Data is normally distributed.
- **H_1 :** Data is not normally distributed.

Decision:

- $p < 0.05 \rightarrow$ reject $H_0 \rightarrow$ evidence of non-normality
- $p \geq 0.05 \rightarrow$ fail to reject $H_0 \rightarrow$ not enough evidence of non-normality

In our project

We used Shapiro-Wilk for the **smoker and non-smoker groups** before deciding which test to use.

Both groups had very small p-values, so we rejected normality for both groups.

Therefore, we used the **Mann-Whitney U test**.

4. Levene's Test

Levene's test checks whether two groups have **similar variances**.

In simple words:

It checks whether the spread/variation of the two groups is similar.

Hypotheses:

- H_0 : The variances are equal.
- H_1 : The variances are not equal.

If $p < 0.05$, we reject H_0 and say the variances are significantly different.

Example

Suppose:

- Group A has values spread between 10–20
- Group B has values spread between 10–100

Their spreads are very different, so equal variance may not be reasonable.

5. Mann-Whitney U Test

Mann-Whitney U is used to compare **two independent groups**.

It is especially useful when the data is **not normally distributed**.

In our project

We compared:

Smoker charges vs Non-smoker charges

Since the normality assumption was not satisfied, we used Mann-Whitney U instead of a t-test.

Our p-value was extremely small:

$$p \approx 5.27 \times 10^{-130}$$

Since $p < 0.05$:

Reject H_0

So, there is a significant difference between the medical charges of smokers and non-smokers.

6. One-Way ANOVA

ANOVA stands for **Analysis of Variance**.

One-Way ANOVA is used when we want to compare the means of **three or more groups**.

In our project

We compared:

Medical charges across different regions

There are four regions:

- Northeast
- Northwest
- Southeast
- Southwest

Hypotheses:

- H_0 : All group means are equal.
- H_1 : At least one group mean is different.

Our result:

$p = 0.03089$

Since:

$0.03089 < 0.05$

we reject H_0 .

Therefore, there is evidence that the average medical charges are not the same across all regions.

ANOVA tells us that a difference exists, but not exactly which two regions are different.

7. Multiple Linear Regression

Multiple Linear Regression is used when we want to predict one numerical value using **multiple input variables**.

In our project:

Target: Medical charges

Inputs:

- Age

- BMI
- Children
- Sex
- Smoker
- Region

The basic idea is:

We use several factors together to predict medical charges.

For example, instead of predicting charges using only age, we use age + BMI + smoker status + children + region + sex.

This gives us a more complete model.

8. OLS

OLS stands for:

Ordinary Least Squares

OLS is the method we use to find the best coefficients for our linear regression model.

It tries to make the prediction errors as small as possible.

The error for one observation is called a **residual**.

$$\text{Residual} = \text{Actual} - \text{Predicted}$$

Example

Actual charge = ₹20,000

Predicted charge = ₹18,000

Residual:

$$20,000 - 18,000 = 2,000$$

OLS tries to choose the regression coefficients so that the **squared residual errors are as small as possible overall**.

In our code:

```
model = sm.OLS(y, X).fit()
```

9. Regression Coefficients

A coefficient tells us how the predicted target changes when one input changes, **while keeping the other inputs constant**.

Example from our project

The coefficient for age is approximately:

256.98

So, for a one-year increase in age, predicted charges increase by about **\$256.98**, assuming the other variables stay the same.

The smoker coefficient is approximately:

23,859.79

So being a smoker is associated with about **\$23,860 higher predicted charges** compared with the reference category, while keeping the other variables constant.

10. Regression p-value

A regression coefficient also has a p-value.

It helps us decide whether that particular predictor has a statistically significant relationship with the target.

Simple rule:

- **$p < 0.05 \rightarrow$ significant**
- **$p \geq 0.05 \rightarrow$ not significant**

Example

For smoker_yes:

p-value is extremely small.

Therefore, smoker status is statistically significant in our model.

For sex_male:

$p \approx 0.69$

Since $0.69 > 0.05$, sex is not statistically significant in our model.

11. R^2

R^2 tells us how much of the variation in the target is explained by our model.

Our model has:

$$R^2 = 0.751$$

This means that approximately **75.1% of the variation in medical charges is explained by the predictors in our model.**

Higher R^2 generally means the model explains more variation, but a high R^2 alone does not mean that the model is perfect.

12. Adjusted R^2

Adjusted R^2 is similar to R^2 , but it also considers the **number of predictors** in the model.

This is useful because adding more variables can increase ordinary R^2 even if those variables are not very useful.

Our model:

$$R^2 = 0.751$$

$$\text{Adjusted } R^2 = 0.749$$

Since they are very close, the model's explanatory power is not being greatly inflated by the number of predictors.

13. Residual

A residual is simply:

Actual value – Predicted value

$$e = y - \hat{y}$$

If:

Actual = 15,000

Predicted = 13,000

then:

Residual = 2,000

If:

Actual = 13,000

Predicted = 15,000

then:

Residual = -2,000

Residuals help us check whether our regression model is behaving properly.

14. Residual vs Fitted Plot

This plot shows:

Residuals vs Predicted values

We use it to look for patterns in the errors.

Ideally, the points should be:

- randomly scattered
- around zero
- without a clear pattern

If we see a pattern or the spread keeps increasing/decreasing, it can indicate a problem with the model.

One important problem we can detect is **heteroscedasticity**.

15. Q-Q Plot

Q-Q means:

Quantile-Quantile

A Q-Q plot helps us check whether the residuals are approximately normally distributed.

If the points are close to the diagonal line:

→ residuals are reasonably close to normal.

If the points strongly move away from the line:

→ residuals may not be normally distributed.

So remember:

Q-Q plot → visual check for residual normality

16. Jarque-Bera Test

Jarque-Bera is another test used to check whether the **residuals are normally distributed**.

It mainly looks at:

- **Skewness**
- **Kurtosis**

Hypotheses:

- H_0 : Residuals are normally distributed.
- H_1 : Residuals are not normally distributed.

If $p < 0.05$:

→ Reject H_0

→ Evidence that residuals are not normally distributed.

Important difference

Shapiro-Wilk: used in our hypothesis testing to check the groups.

Jarque-Bera: used as part of our regression diagnostics to check the residuals.

17. Breusch-Pagan Test

Breusch-Pagan checks for **heteroscedasticity**.

First understand:

Homoscedasticity

The error spread is roughly constant.

Heteroscedasticity

The error spread changes as the predicted value changes.

The hypotheses are:

- H_0 : Constant variance
- H_1 : Non-constant variance

Our p-value was approximately:

$$1.30 \times 10^{-22}$$

This is much smaller than 0.05.

Therefore:

Reject $H_0 \rightarrow$ Heteroscedasticity is present.

Because of this, we used **HC3 robust standard errors**.

18. Multicollinearity

Multicollinearity happens when two or more independent variables are strongly related to each other.

Simple example

Suppose we use:

- Age in years
- Age in months

These two variables contain almost the same information.

The model may have difficulty deciding which variable should get the effect.

This can cause:

- unstable coefficients
 - larger standard errors
 - difficulty interpreting individual variables
-

19. VIF

VIF stands for:

Variance Inflation Factor

VIF is used to check **multicollinearity**.

Simple interpretation:

- $VIF \approx 1 \rightarrow$ very little multicollinearity
- VIF below 5 $\rightarrow$ generally acceptable
- VIF above 5 $\rightarrow$ possible concern
- Very high VIF $\rightarrow$ serious multicollinearity

In our project, the highest VIF was only around **1.65**.

Therefore:

Our model does not have serious multicollinearity.

Easy way to remember

VIF → checks whether predictors are too related to each other.

20. HC3 Robust Standard Errors

HC3 stands for:

Heteroscedasticity-Consistent standard errors, version 3

We use HC3 when heteroscedasticity is present.

Important point:

HC3 does not change the regression coefficients.

It changes the:

- Standard errors
- t-statistics
- p-values
- Confidence intervals

So, the estimated effect of smoking remains approximately **23,859.79**, but the uncertainty around that estimate is calculated more safely using HC3.